Sitzungsprotokoll – 4 November 2026

Clifford Seminar Group, Universidad Distrital

Topic: Thesenpapier 01, Problem 3 (dual space and bidual), continued from last session.
At the board: Julián. **Protocol:** Sebastián Rodríguez.
Present: Adriano, Camilo, David, Jaime, José, Juanes, Laura, Manu, Sebastián; online: Alexander, Nuri.

Recap

We started with a short recap for those who missed last session. The elements of V^* are linear maps that take a vector and give a number, so we can think of them as measuring devices. Going back to the Kepler plane, we took the transverse sensor $f_2(r,\theta) = r\cos\theta$, which measures the projection of the position onto a chosen axis. Its differential is

$$df_2 = \cos\theta\, dr - r\sin\theta\, d\theta.$$

(It was first written with a +; someone caught the sign.) We also recalled that $dr(e_r) = 1$, $dr(e_\theta) = 0$, and so on, i.e. $\langle \omega^i, e_j\rangle = \delta^i_j$.

Problem 3(a): is there a canonical map $V \to V^*$?

Take two bases $\mathcal{E} = \{e_i\}$ and $\widetilde{\mathcal{E}} = \{\tilde e_i\}$ with $\tilde e_i = T(e_i)$, and define the dual map T^* by $\langle T^*\alpha, v\rangle = \langle \alpha, Tv\rangle$, i.e. $T^*\alpha = \alpha \circ T$.

Julián asked what we need for T^* to exist. After some back and forth, the answer was: only that T be linear, because then $\alpha \circ T$ is again linear and lives in V^*. Invertibility is not needed here; we only ask for it because T has to send a basis to a basis. (Side comment: in quantum mechanics the existence of the adjoint is more delicate; the example given was the momentum operator.)

Inserting $T^{-1}T$ gives $(T^{-1})^* = (T^*)^{-1}$. Then, from $\langle \tilde\omega^j, Te_i\rangle = \delta^j_i$, we get $T^*\tilde\omega^j = \omega^j$, so

$$\tilde e_i = T(e_i), \qquad \tilde\omega^i = (T^{-1})^*\omega^i.$$

Julián made the point that this is the real reason behind the "covariant vs. contravariant" story: the dual basis transforms with the inverse because we impose $\langle \tilde\omega^j, \tilde e_i\rangle = \delta^j_i$ in every basis, not because of some extra rule.

Now let $\phi_{\mathcal{E}} \colon V \to V^*$, $\phi_{\mathcal{E}}(e_i) = \omega^i$ (and similarly $\phi_{\widetilde{\mathcal{E}}}$). From the above,

$$\phi_{\widetilde{\mathcal{E}}} = (T^*)^{-1} \circ \phi_{\mathcal{E}} \circ T^{-1},$$

which always holds. (Julián's remark here: the diagram only tells you that both paths have the same start and end, not that they are equal.) The question is whether $\phi_{\widetilde{\mathcal{E}}} = \phi_{\mathcal{E}}$ for every T. Someone proposed $T = \mathrm{id}$, which works trivially but misses the point, since the claim has to hold for all T. If we assume it, we get $\phi_{\mathcal{E}} = T^* \circ \phi_{\mathcal{E}} \circ T$, that is,

$$b(u,v) := \langle \phi_{\mathcal{E}}(u), v\rangle \quad \text{must satisfy} \quad b(Tu,Tv) = b(u,v).$$

So the two maps agree only when T is an isometry of b. Julián stressed calling b a symmetric bilinear form and not an inner product, since we never ask for positivity; later it will play the role of the metric. Rotations, reflections and translations were given as examples of isometries.[1]

[1] Small correction: translations are not linear maps of V, so only rotations and reflections count here.

Example (dilatation). Take $T = \lambda\,\mathrm{id}$ with $\lambda \neq 0$ (so it is invertible) and $\lambda \neq \pm 1$ (-1 is a reflection, which is an isometry). Then $T^* = \lambda\,\mathrm{id}$ and

$$T^* \circ \phi_{\mathcal{E}} \circ T = \lambda^2 \phi_{\mathcal{E}} \neq \phi_{\mathcal{E}}.$$

Someone asked whether the λ could be "absorbed" into α some other way, as with complex conjugation. No: we work over $\mathbb{R}$, and in any case this is just evaluating a functional, not a complex inner product.

Conclusion: the map $V \to V^*$ depends on the basis, so there is no canonical isomorphism. Julián added a comment on matrices: a bare matrix does not tell you which basis it is in, or even whether it is a vector, a covector or a tensor, so it hides exactly what we were discussing.

Problem 3(b): the bidual

Define $\mathrm{ev}\colon V \to V^{**}$ by $\mathrm{ev}_v(\alpha) = \alpha(v)$. There is no basis anywhere in the definition. Using the dual property twice,

$$\langle T^{**}\,\mathrm{ev}_v, \alpha\rangle = \langle \mathrm{ev}_v, T^*\alpha\rangle = \langle \alpha, Tv\rangle = \langle \mathrm{ev}_{Tv}, \alpha\rangle \quad\Longrightarrow\quad T^{**} \circ \mathrm{ev} = \mathrm{ev} \circ T$$

for every linear T, not just isometries. (Someone asked whether T^{**} means applying T twice. No: it is a single map, the dual of T^*.) So ev is canonical. Since V is finite-dimensional, it is an isomorphism and $T^{**} = T$. This part took a while: a lot of the confusion was between ev (a map $V \to V^{**}$) and ev_v (a map $V^* \to \mathbb{R}$).

We also discussed what happens in infinite dimensions (example: polynomials). There $\mathrm{ev}(V)$ is only a subspace of V^{**}.[2]

Conclusions and next session

Before the conclusions, Julián noted that Problem 3 took about a session and a half and that we did not get to quadratic spaces. He is working with the director on a more realistic plan. Then we went over the conclusions together:

1. The main takeaway: the dual basis transforms with $(T^{-1})^*$ as a consequence of the duality relation.
2. There is no canonical $V \cong V^*$ (it only works for isometries), while $V \cong V^{**}$ is canonical. The bidual is more of a bonus.
3. Why this matters physically: velocities live in V and momenta in V^*, so $\vec{p} = m\vec{v}$ cannot be literally true. We need extra structure (a bilinear form) to go from one to the other. That is where $b^\flat$ and $b^\sharp$ come in.

For next time: try Problem 4 ($p = m\,b^\flat(v)$, $b^\sharp$ and $H = \frac{1}{2m} b^*(p,p)$).

[2] ev is still canonical in infinite dimensions; what fails is that it is no longer onto.